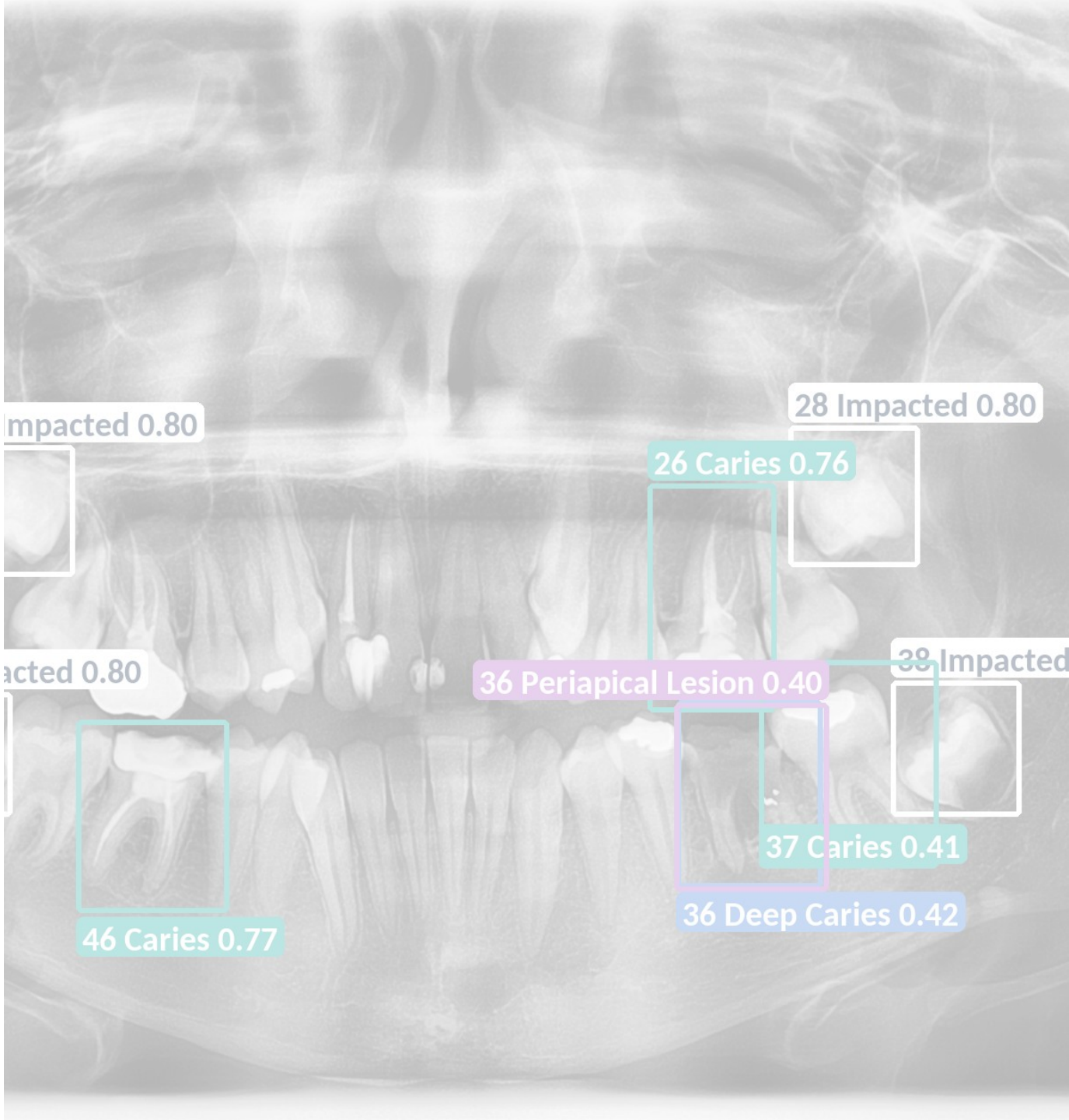
DentalAI

A second opinion on every panoramic X-ray — the dentist stays in charge

Team: Seif Ahmed · Abdullah Tarek Habashy · 3 team names: UNVERIFIED
Alexandria University (CDS) · Advisor: Ashraf Eshta
IEEE IC-FTAI 2026 · Student Project Track



Background: shipped model on a public DENTEX image (CC BY-NC-SA)

The problem

Findings get missed

Easy to overlook on a busy day.

History is scattered

Paper files, separate software.

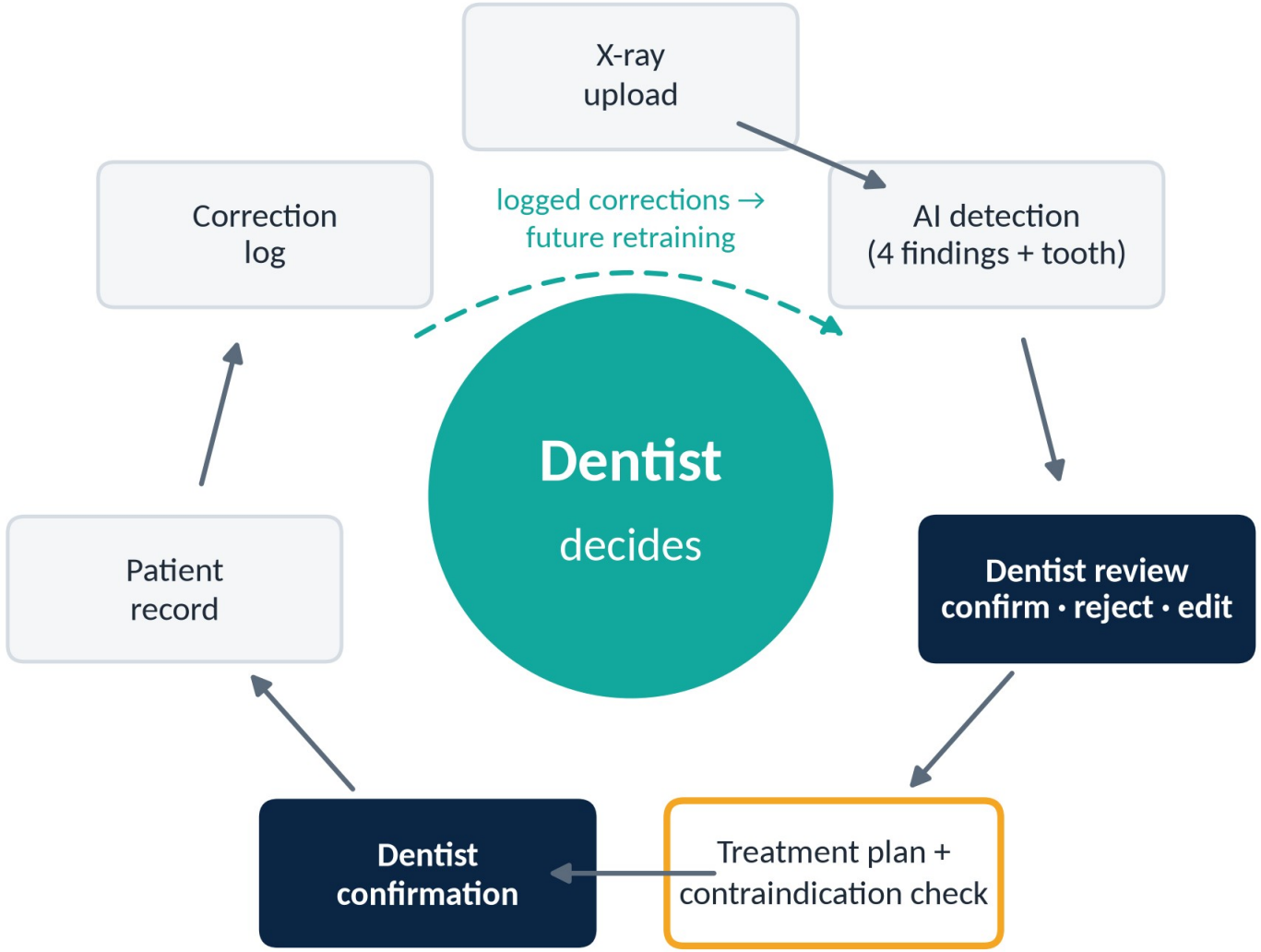
Drug safety from memory

Allergies checked by recall.

AI ignores the dentist

Corrections are simply lost.

Our solution: the closed loop



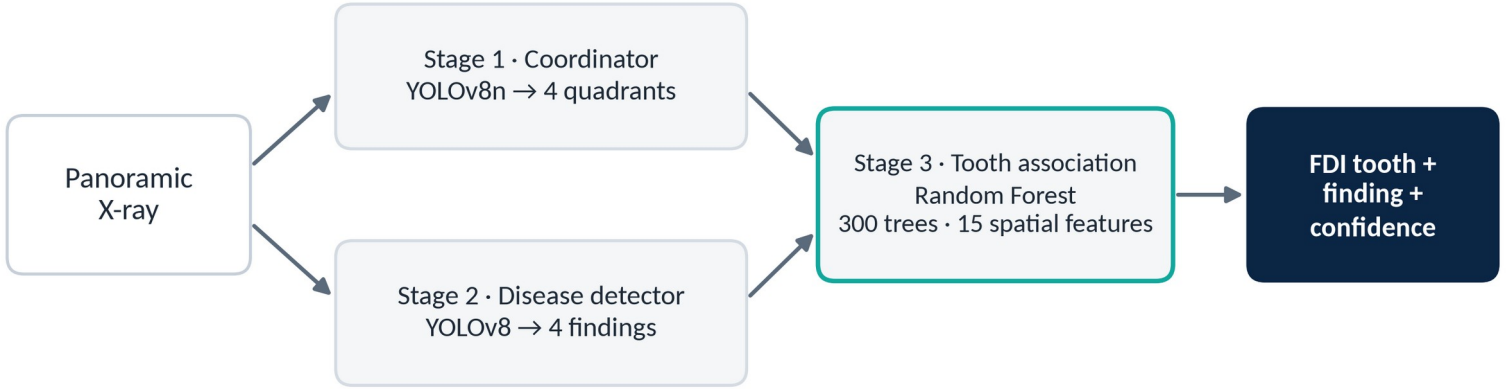
Others detect.

DentalAI connects detection, decision, treatment and record.

Every correction is logged as future training data.

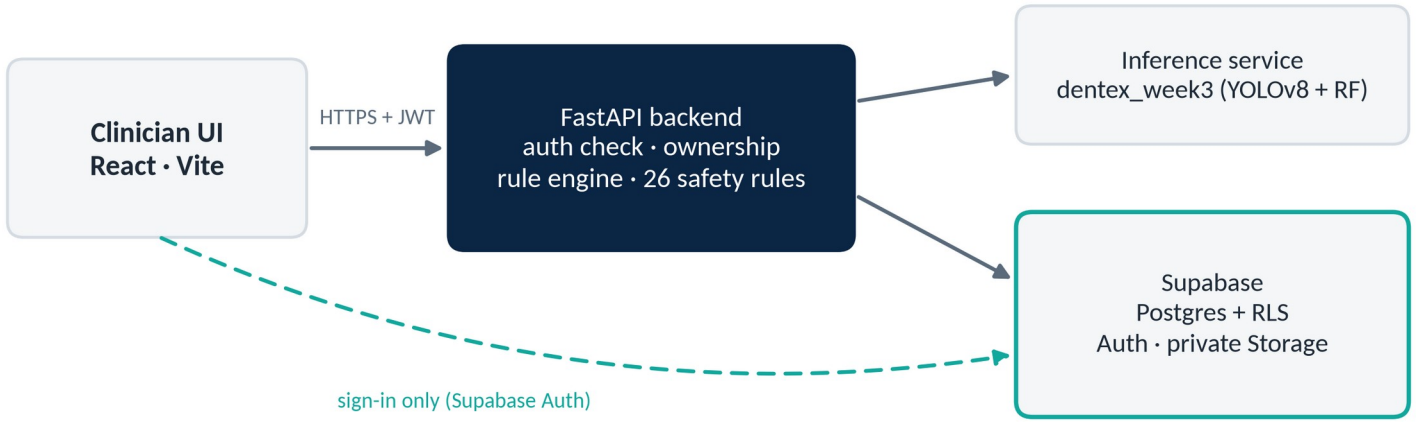
How it works

Model pipeline



Findings: caries · deep caries · periapical lesion · impacted tooth

System



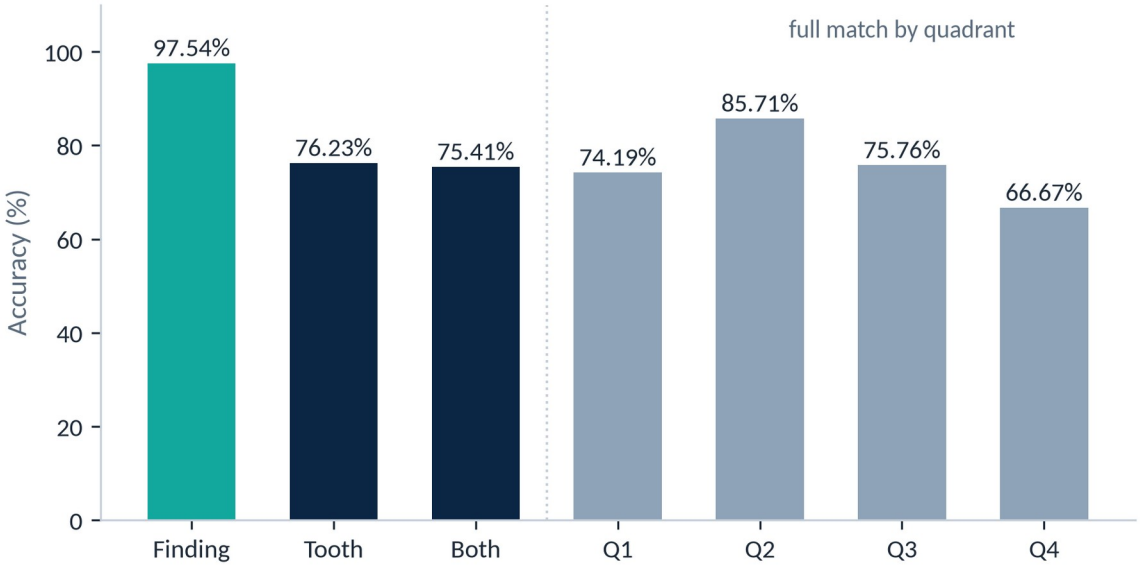
Results

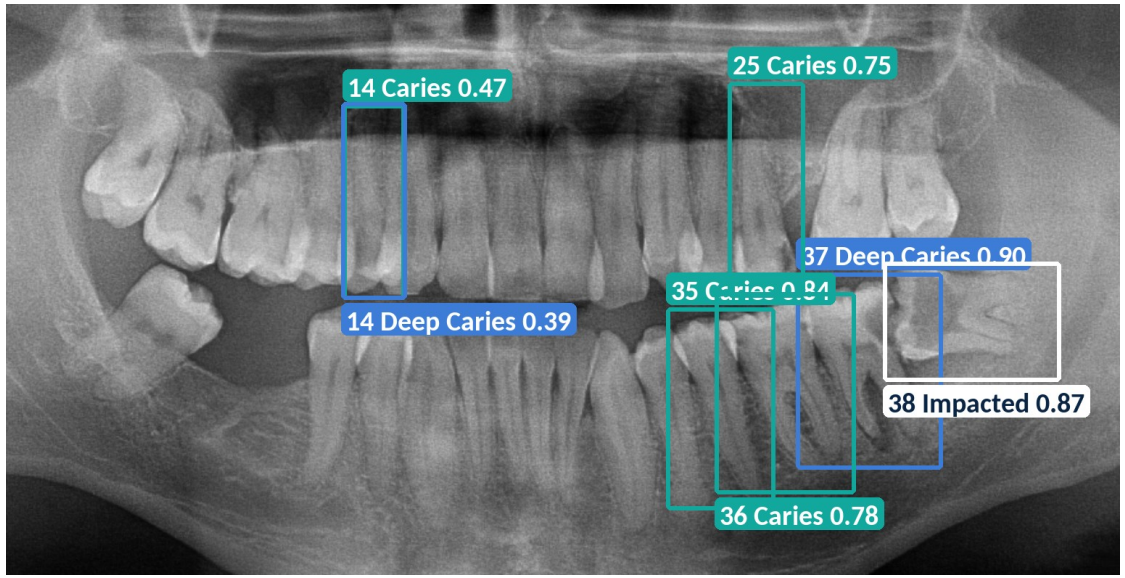
Shipped pipeline · held-out benchmark · 23 images, 122 detections

97.54%
finding label correct

76.23%
tooth number correct

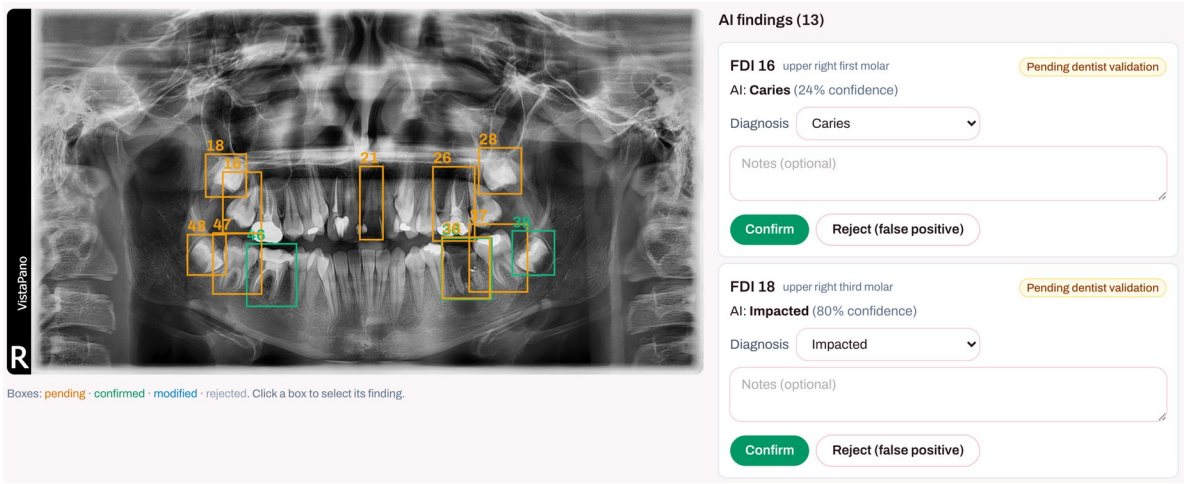
75.41%
finding + tooth correct



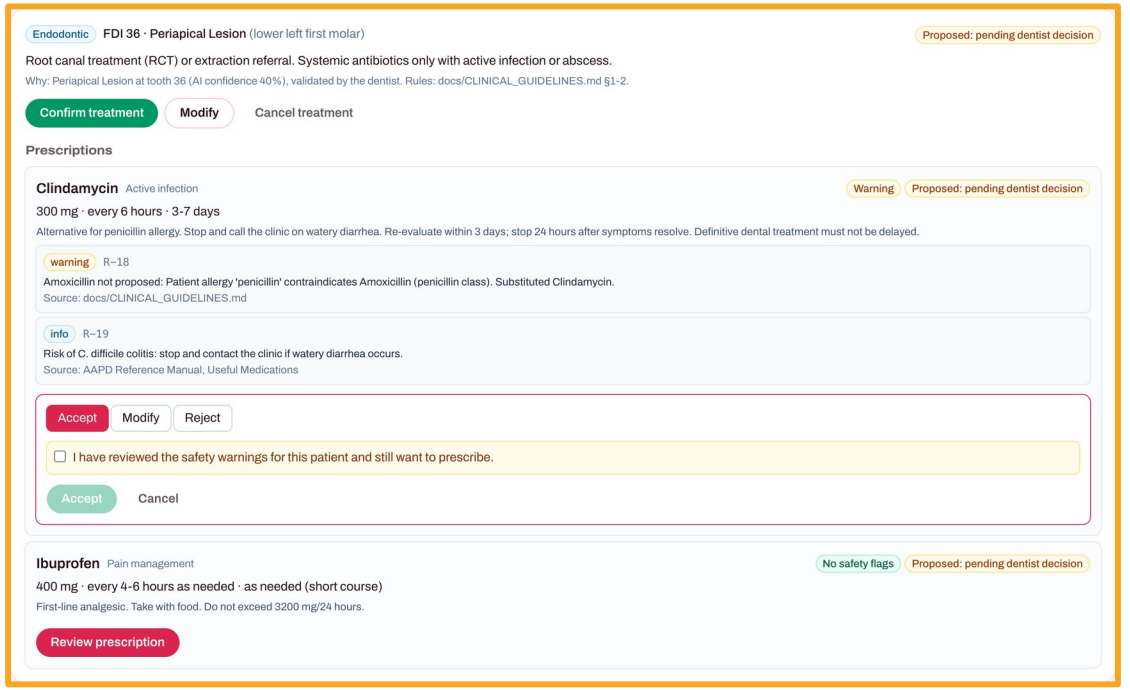


Public DENTEX image · detections ≥ 0.35 shown

In the clinic: safety by design



Findings stay pending until the dentist acts.



Penicillin allergy: warning must be acknowledged.

Caries

No drug

Infection

Drug, no conflict

Allergy

Drug with conflict

AI proposes. Dentist decides.

Impact & next steps

Done

Full loop, end to end

Login + per-dentist access

196 automated tests in CI

Next

Clinic pilot

Clinician rule review

Retraining from corrections

PDPL 151/2020 compliance

Code & docs on GitHub

Thank you — questions welcome.

Backup — Q&A

Competition · Data · Limitations · Privacy · Testing · Clinical rules

Backup · Competitive landscape

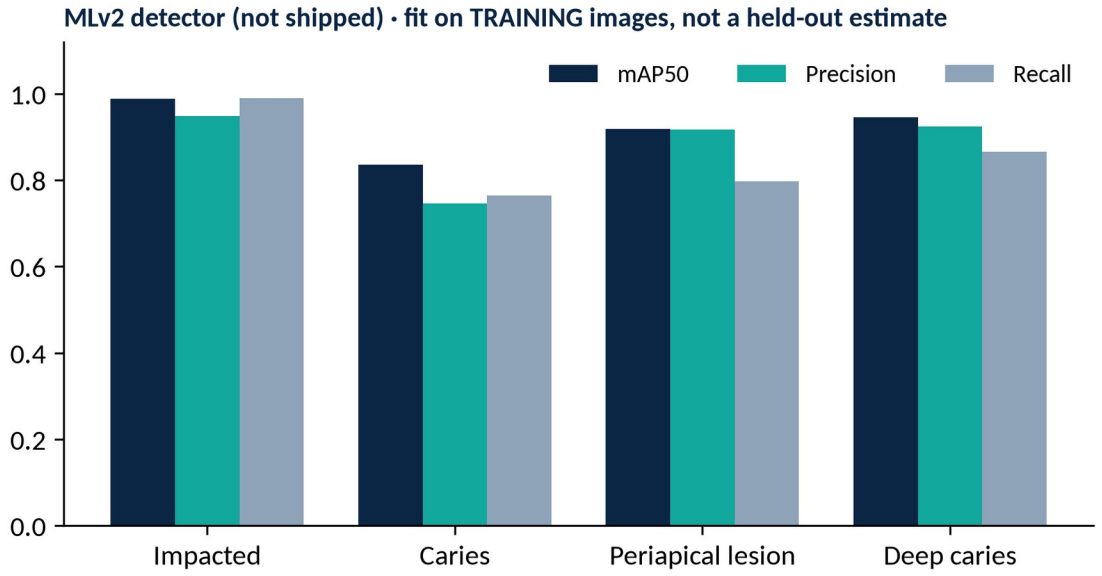
Product	Detects findings on X-rays	Dentist confirms each finding	Treatment + drug-safety check	Longitudinal record	Corrections logged
Diagnocat	Yes	verify	verify	verify	verify
Pearl	Yes	verify	verify	verify	verify
Overjet	Yes	verify	verify	verify	verify
dentalXrai	Yes	verify	verify	verify	verify
DentalRx	verify	verify	verify	verify	verify
DentalAI	Yes (4 findings + tooth)	Yes, required	Yes, 26 rules, gated	Yes	Yes (retraining planned)

Competitor cells are UNVERIFIED: check each vendor's current public product pages before presenting.

Our claim is narrow: DentalAI links detection to the dentist's decision, treatment safety and the patient record in one loop.

Backup · Data & training

Data: DENTEX 2023 panoramic X-rays (public, CC BY-NC-SA). Tufts + Loki model set aside; Roboflow never used.
Classes: caries, deep caries, periapical lesion, impacted; FDI tooth numbers.
Training (Kaggle, 2× T4): YOLOv8n coordinator, 50 epochs, 640 px; YOLOv8s detector (MLv2), 50 epochs, 640 px; Ultralytics default augmentation.
Tooth association: Random Forest, 300 trees, 15 spatial features, balanced classes.
Held-out test: 23 images, 122 detections (results slide).
Shipped detector checkpoint differs from the MLv2 run: its training split is UNVERIFIED.



MLv2 validated on its own training folder (val = train), so these numbers show fit, not generalization. We report only the held-out benchmark as performance.

Backup · Limitations & honest risks

Small test set: 23 held-out images, 122 detections; no held-out mAP or recall for the shipped detector yet.

Tooth numbering: weakest output (76.23%). Older enumerators shifted numbers when a tooth was missing; the Random Forest was added to fix this.

Uncalibrated confidence: scores are model probabilities, not clinical probabilities.

Domain gap: trained and tested on public DENTEX images; real clinic X-rays may differ.

Feedback loop: corrections are logged; no retraining has been run yet.

Clinical rules: source-backed (ADA, AHA, AAPD) but not yet clinician-validated; no clinic pilot yet.

Backup · Privacy & compliance

Synthetic data only: demo patients are synthetic QA profiles; X-rays are public DENTEX images.

Access: every API call needs a signed-in dentist; each dentist sees only their own patients (others answer 404).

Database: Supabase row-level security with no client access; only the backend reads and writes.

Images: private storage bucket; time-limited signed links (1 hour).

Audit: every correction and override is recorded with the dentist's id.

Egypt PDPL 151/2020: compliance review planned before any clinic pilot (consent, data residency, retention).

Backup · Testing & QA evidence

171	25	3	6
backend tests	frontend tests	CI jobs	synthetic profiles

CI: repo guard (no weights, secrets or keys committed), backend (compile + tests), frontend (typecheck + tests + build).

Three demo branches tested: caries → no drug; periapical lesion + infection → amoxicillin; penicillin allergy → clindamycin with warning.

HTTP tests: sign-in required (401), cross-dentist access blocked (404), decision gates enforced (422).

Live runs: full loop on a real Supabase project with throwaway accounts, deleted afterwards.

Backup · Clinical rules

Pending clinician validation

Finding	Proposed treatment	Drugs
Caries	Restorative referral	Never antibiotics
Deep caries	Pulp capping / endodontic evaluation	Only if infection reported
Periapical lesion	Root canal or extraction referral	Only if infection reported
Impacted tooth	Oral surgery evaluation	Prophylaxis only for cardiac risk

26 safety rules, for example:

R-01 penicillin allergy → no amoxicillin

R-02 ibuprofen + anticoagulant

R-14 no clindamycin for prophylaxis (AHA 2021)

R-23 daily-dose limits

R-18 substitute shown with reason

Warning → acknowledge · Contraindicated → written reason

Sources: ADA 2019, AHA 2021, AAPD Reference Manual, AGS Beers 2023. Antibiotics are proposed only when the dentist reports infection (antibiotic stewardship).